

# 1 Motivation

There is anecdotal evidence that college students are moving away from fields exposed to AI capabilities [computer science majors](#), and [radiologists](#).

Expectations about demand for a profession, which turn out to be wrong ex post, can lead to shortages or gluts in a profession.

For example, in 2016, AI researcher Geoffrey Hinton, who would go on to win the Nobel Prize in 2024, said “[people should stop training radiologists now](#)”, and that “within five years [by 2021] deep learning is going to do better than radiologists.” Radiologists are still employed in 2025 and some companies report a “[supply-demand gap](#)” for radiologists.

## Classes of Questions

- Are AI-exposed education fields suffering a decline or deceleration in enrollment since the launch of ChatGPT? **Evidence today.**
- If so, to what extent is this an ex post reaction to declines in hiring/wages vs. *ex ante expectations* about future automation risk?
  - For example, we might find that the slowdown in computer science enrollment is a reaction to labor market deterioration, whereas the shortage in radiology was induced by medical students moving away from a field they expected to be automated.
- Most speculatively, can we predict which professions might have shortages or gluts if automation ends up being less of a big deal than we thought? This might be viewed as the labor market side of an AI bubble.

This idea is in the air among economists: [\[Brynjolfsson et al., 2025\]](#) released a paper on declining headcount of early career software developers and customer service employees since ChatGPT was launched. In the conclusion, they speculate that “The adoption of new technologies typically leads to heterogeneous effects across workers, resulting in an adjustment period as workers reallocate from displaced forms of work to new forms with growing labor demand (Autor et al., 2024). **Such endogenous adjustment may already be happening with AI, with emerging evidence of shifts in college majors away from AI-exposed categories such as computer science.**”

# 2 Literature

## 2.1 College Major Choice Broadly

Altonji et al Handbook chapter (2016) Missing Manual, Dynarski et al, NBER WP (2013)

## 2.2 AI and Jobs

Acemoglu/Restrepo Tasks frameworks, and earlier. Felten et al, 2021.  
Felten et al, 2023.

Timmerman, 2025  
[Source](#).

### 2.2.1 AIs in Medicine

Combining Human Expertise with Artificial Intelligence: Experimental Evidence from Radiology

## 3 National Student Clearinghouse (NSC)

This is the data I use on college major enrollments. It is self-reported by colleges representing 97% of enrollment. Colleges participate in NSC voluntarily to save administrative costs by outsourcing thousands of enrollment and degree verification requests to loan servicers, employers, and other entities. Accuracy is incentivized because errors create immediate operational problems (students incorrectly placed in loan repayment, failed background checks) that generate complaints and manual work, eliminating the cost savings.

The publicly available data are reported 45 days after start and finish of school semesters, at the state-major level. For all of my analyses, I am aggregating to the major level.

## 4 Analyses

### 4.1 Caveat on data

### 4.2 A change in enrollment from $t$ to $t + 1$

One issue with the data currently, is it is a **stock** of people across all years majoring in each area. I am primarily interested in **flows**, though, i.e. people declaring a major in year  $t$ .

Consider a decline in the total people majoring in Computer Science, I cannot distinguish between the following.

- Existing majors: People in that major could have left Computer Science (and either switched to another major or dropped out)
- More people graduated with that degree than declared that major
  - This could be due to overall trends in college attendance
  - Or due to major-specific trends.

Micro data would allow me to disentangle these, however. This is purchasable for less than \$1000, but I need to define the student target population (and provide names).

### 4.3 How to define AI-exposed majors.

We calculate a static measure of AI exposure by college major. Doing so requires a way to map from occupations, where past researchers have quantified AI exposure, to majors.

**Table 1:** Construction of Major-Level AI Exposure Scores

| Step                                           | Description                                                                                                                                                                                                                                                                                                                              |
|------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <i>Data Sources</i>                            |                                                                                                                                                                                                                                                                                                                                          |
| (1)                                            | <b>Occupation-level AI exposure:</b> Felten et al. (2023) Language Modeling AIOE scores for 774 SOC occupations                                                                                                                                                                                                                          |
| (2)                                            | <b>Occupation-major linkage:</b> IPUMS ACS 2013-2017 (5-year estimates) for ages 22-35, linking field of degree to current occupation. We use this vintage to ensure SOC code definitions align with those in Felten et al. (2023). <sup>1</sup>                                                                                         |
| (3)                                            | <b>Enrollment data:</b> National Student Clearinghouse 2023 undergraduate enrollment by 4-digit CIP code                                                                                                                                                                                                                                 |
| <i>Construction of Major-Level AI Exposure</i> |                                                                                                                                                                                                                                                                                                                                          |
| (1)                                            | <b>Link majors to occupations:</b> For each major $c$ , identify all ACS individuals whose field of degree maps to that major                                                                                                                                                                                                            |
| (2)                                            | <b>Assign occupation-level exposure:</b> Each individual $i$ receives the AI exposure score of their occupation: $\text{AIOE}_{\text{occ}(i)}$ from Felten et al. 2023.                                                                                                                                                                  |
| (3)                                            | <b>Aggregate to major level:</b> Calculate major-level exposure as the weighted average AIOE across all occupations that major’s graduates enter:<br>$\text{AI\_Exposure}_c = \frac{\sum_{i \in c} w_{i,c} \times \text{AIOE}_{\text{occ}(i)}}{\sum_{i \in c} w_{i,c}}$ where $w_{i,c}$ is the weight on person $i$ used for major $c$ . |

On calculation of  $w_{i,c}$ :

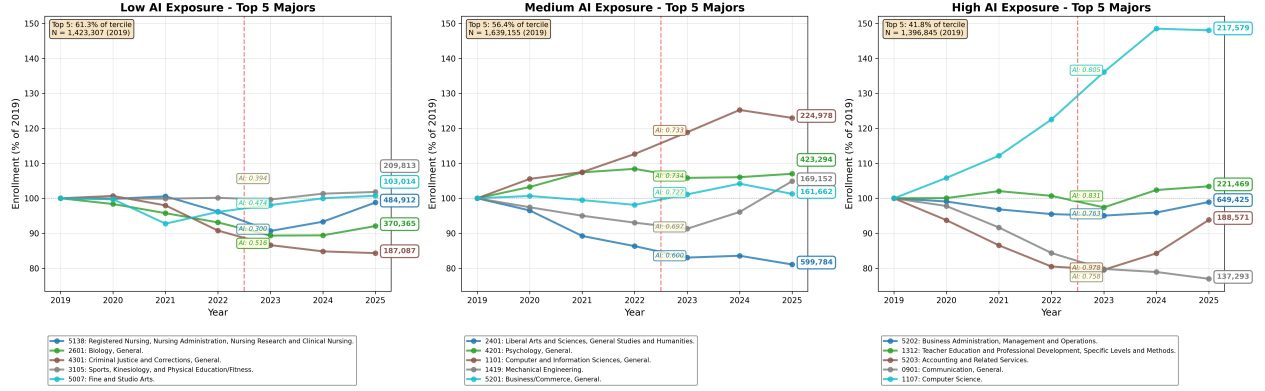
Since ACS uses FOD (Field of Degree) codes while enrollment data uses CIP (4-digit Classification of Instructional Programs) codes, I harmonize using a many-to-many NCES crosswalk, and then (ii) weight by empirical enrollment shares in each CIP when an ACS FOD maps to many CIPs.

For example, if a FOD maps to multiple CIPs, which we denote  $\text{CIP}(\text{FOD}_i)$ , then we split that person using empirical enrollment shares in each of those CIPs, from 2023.  $\text{PERWT}_i$  is the person weight from ACS.

$$w_{i,c} = \text{PERWT}_i \times \frac{\text{Enrollment}_c}{\sum_{c' \in \text{CIP}(\text{FOD}_i)} \text{Enrollment}_{c'}}$$

#### 4.3.1 Is AI exposure the measure we want?

The Felten et al measure is **not** designed to capture automation risk, as they say, “we remain agnostic to whether occupational exposure to generative AI is likely to result in automation



**Figure 1:** Enrollment by AI occupational exposure over time. Data from NSC.

vs. augmentation of occupations”.

## 4.4 Descriptives

### 4.4.1 Quantities of enrollment

We see a deceleration in computer science and computational and information science between 2024 and 2025, as reported on in the popular press.

But, other trends, unrelated to generative AI, have also been occurring. For example, the secular decline in Communication majors (high AI exposure) and Criminal Justice (low AI exposure) majors, which has been occurring since at least 2019.

### 4.4.2 Wages by AI exposure

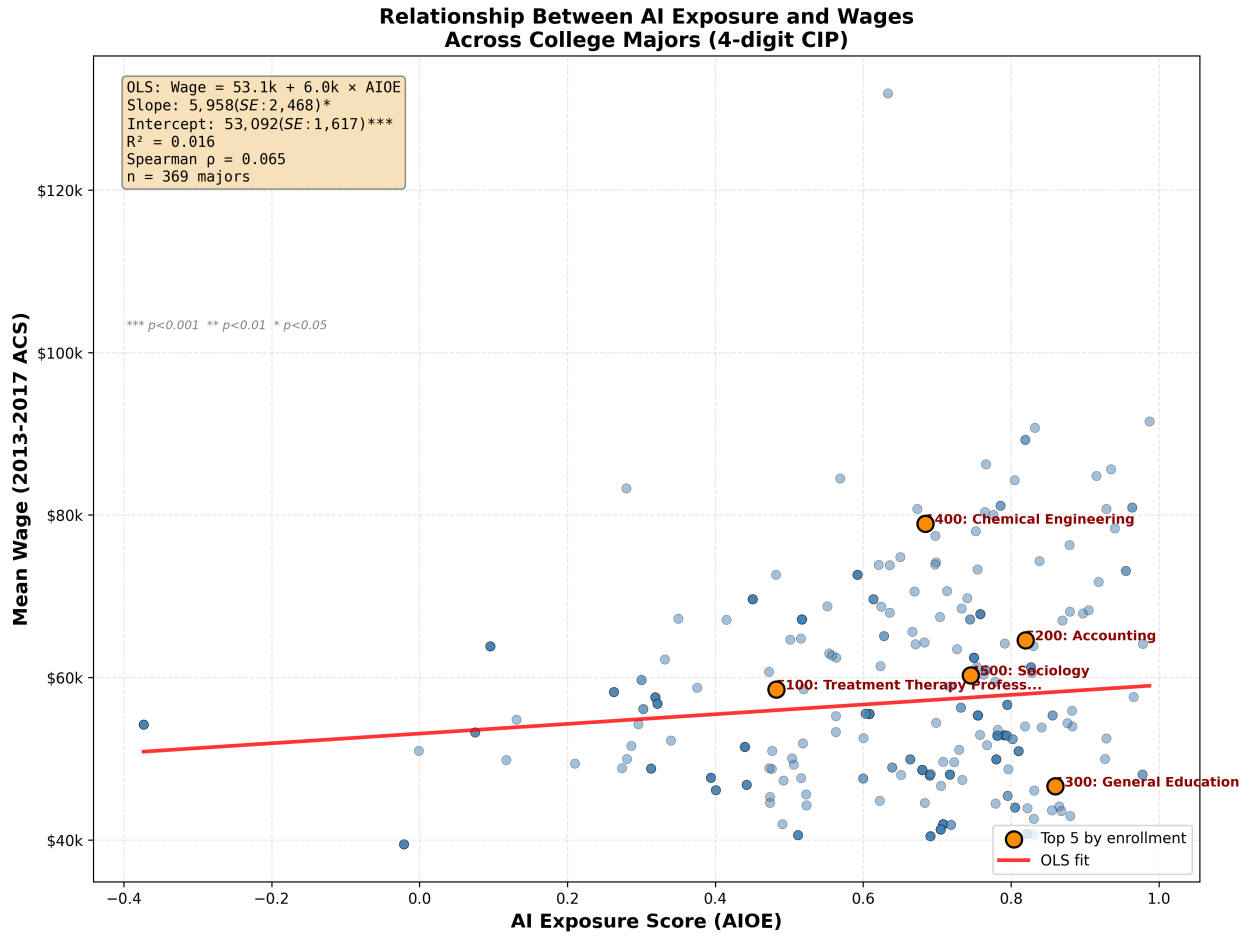
The most recent wage  $\times$  major data which are compatible with the above SOC codes is the 2013-17 ACS data.

## 4.5 Motivating DiDs

It seems like if anything was happening, it was not happening until 2025. So I run a DiD with just two periods between 2024-5. I cut the data into high and low AI exposure majors (above/below median AIOE), and test for differential changes in enrollment levels and enrollment growth rates.

This specification is not well-motivated by any specific exogenous shock in 2024/2025. Most papers that try to do DiDs like this use the launch of Chat GPT on Nov 2022 as the shock, but it’s pretty clear from the raw data we should not expect to see anything there.

Table 2 presents a simple difference-in-differences comparing 2024-2025 enrollment changes for high vs. low AI exposure majors. Columns (1)-(2) examine level changes ( $\text{Enrollment}_{2025} - \text{Enrollment}_{2024}$ ), while columns (3)-(4) examine growth acceleration (whether enrollment growth accelerated or decelerated relative to the prior year). Columns (2) and (4) add controls for log mean wages (2013-2017).



**Figure 2:** Wage vs. AI Occupational Exposure by Major. Each point represents a 4-digit CIP code. Wages are mean earnings from the 2013-2017 ACS for degree holders ages 22-35. AIOE scores are derived from Felten et al. (2023). Includes OLS regression line with slope, intercept, p-values, and R<sup>2</sup>. Top 5 most populous majors are labeled.

High AI exposure majors show slightly higher level growth (+297-302 students,  $p > 0.10$ ) but marginally significant growth *acceleration* (+3.8 percentage points,  $p < 0.10$ ). With only one “post-period” and wide standard errors, these results should be interpreted cautiously.

**Table 2:** Difference-in-Differences: AI Exposure and Enrollment Changes (2024-2025)

|                    | Level Change<br>(Students) |                     | Growth Acceleration<br>(Percentage Points) |                 |
|--------------------|----------------------------|---------------------|--------------------------------------------|-----------------|
|                    | (1)                        | (2)                 | (3)                                        | (4)             |
| High AI Exposure   | 297<br>(450)               | 302<br>(445)        | 3.81*<br>(2.20)                            | 3.80*<br>(2.20) |
| Log Wage (2013-17) | –                          | 2,924***<br>(1,090) | –                                          | -6.23<br>(5.36) |
| Observations       | 260                        | 260                 | 256                                        | 256             |
| R-squared          | 0.002                      | 0.029               | 0.012                                      | 0.017           |
| Wage Control       | No                         | Yes                 | No                                         | Yes             |

Notes: Standard errors in parentheses. High AI Exposure = majors above median AIOE score.

Level change =  $\text{Enrollment}_{2025} - \text{Enrollment}_{2024}$ .

Growth acceleration =  $\text{Growth rate}_{2024-25} - \text{Growth rate}_{2023-24}$ .

Wage control uses log mean wage from 2013-2017 ACS.

\*\*\*  $p < 0.01$ , \*\*  $p < 0.05$ , \*  $p < 0.10$

Table 3 presents an alternative specification using the launch of ChatGPT (November 2022) as the treatment date. I compare average enrollment in the pre-period (2019-2022) to the post-period (2023-2025) for high vs. low AI exposure majors. While level changes show no significant differential (columns 1-2), growth rates reveal that high AI exposure majors grew 7.8 percentage points less than low AI exposure majors ( $p < 0.01$ ). This result is robust to wage controls and suggests that, over the full post-ChatGPT period, enrollment shifted away from AI-exposed fields.

## 5 A simple model of major choice in response to substitutes

The key features we desire in a model are:

- Two majors; roughly, one should be “low-risk”, it pays off the same in each state, and one should be high-risk, it pays off little in the state you get automated in, and a lot otherwise.
- We could think of this as nursing and computer science.
- The risk agents face is macro-risk to their profession, not uncertainty about your own type, which are sometimes studied.

**Table 3:** Difference-in-Differences: AI Exposure and Enrollment Changes (Pre/Post ChatGPT)

|                    | Level Change<br>(Students) |                    | Growth Rate<br>(Percentage Points) |                    |
|--------------------|----------------------------|--------------------|------------------------------------|--------------------|
|                    | (1)                        | (2)                | (3)                                | (4)                |
| High AI Exposure   | 192<br>(1,012)             | 218<br>(1,002)     | -7.83***<br>(2.66)                 | -7.80***<br>(2.66) |
| Log Wage (2013-17) | –                          | 6,219**<br>(2,470) | –                                  | 5.49<br>(6.52)     |
| Observations       | 265                        | 265                | 261                                | 261                |
| R-squared          | 0.000                      | 0.024              | 0.032                              | 0.035              |
| Wage Control       | No                         | Yes                | No                                 | Yes                |

Notes: Standard errors in parentheses. High AI Exposure = majors above median AIOE score.

Pre-period: 2019-2022 (average enrollment). Post-period: 2023-2025 (average enrollment).

Level change =  $\text{Avg Enrollment}_{\text{post}} - \text{Avg Enrollment}_{\text{pre}}$ .

Growth rate =  $(\text{Avg Enrollment}_{\text{post}} - \text{Avg Enrollment}_{\text{pre}}) / \text{Avg Enrollment}_{\text{pre}} \times 100$ .

Wage control uses log mean wage from 2013-2017 ACS.

\*\*\*  $p < 0.01$ , \*\*  $p < 0.05$ , \*  $p < 0.10$

- Expectations about wages for each major.
- A delay between when you decide your major and get your first job (I am not sure this is important)

For now, assume people have equal ability to do any major. And everyone has the same ex-ante beliefs.

So, two majors.  $C$  and  $N$ . We assume one major maps directly into one profession and there is no unemployment. Two potential states of the world, low demand  $L$  (AIs automate computer science jobs) and  $H$  (humans keep those jobs). With probability  $p_H = 1 - p_L$ . We can thus think of  $H$  as a status quo, and  $L$  as a deviation from that status quo.

**Preferences** We assume people have preferences only over their final wage. Conditional on employment they each provide 1 unit of labor inelastically. One way to think about this is people are extremely elastic to their choice of major, but completely inelastically supply labor once school is done.

Furthermore, assume, to start, that wages are only determined by employer demand. And employer demand only varies between the two states,  $H$  and  $L$ .

Demand for each major upon graduation are as follows.

$$D_R(H) > D_S(L) = D_S(H) > D_R(L)$$

For simplicity, we thus denote demand for safe major in either state as  $D_S$

Under risk neutrality, the following indifference in expected wages must hold, although

we have not explained how wages are determined yet.

$$p_H w_R(H) + p_A w_R(A) = w_S(H) \quad (1)$$

## 5.1 Demand

We now quantify the distribution of majors, assume a mass 1 of students choose between the two majors, with  $n_R$  choosing computer science, and  $n_S = 1 - n_R$  choosing nursing.

### 5.1.1 Completely Elastic Demand

If demand for employment in each degree is perfectly elastic at wage  $w_R(H) > w_S > w_R(L)$ , then these wages fully pin down the equilibrium. This means any split of majors, and in fact, any wages, are an equilibrium as long as Equation ?? is satisfied.

### 5.1.2 Partially Inelastic Demand

We now specify demand a little more narrowly, for all three majors we have a linear demand curve of the form

$$D = a - wb$$

And we specify demand for each major as a shift in the demand curve, with the same constant elasticity  $b$ .

$$D_{H,L} = a_H - w_{R,H}b, D_S = a_S - w_Sb, D_{R,L} = a_L - w_{R,L}b$$

Where  $a_H > a_S > a_L$ . On the supply side, wages must satisfy the following indifference condition.

$$E[w_R] = w_S$$

$$p_H w_{R,H} + p_L w_{R,L} = w_S$$

$$p_H \left( \frac{n_R - a_H}{-b} \right) + p_L \left( \frac{n_R - a_L}{-b} \right) = \frac{n_S - a_S}{-b}$$

$$p_H(n_R - a_H) + p_L(n_R - a_L) = (n_S - a_S)$$

$$n_R + p_H(-a_H) + p_L(-a_L) = (n_S - a_S)$$

And quantities must sum to 1.

$$n_R + p_H(-a_H) + p_L(-a_L) = (1 - n_R - a_S)$$

Denoting  $E[a]$  as expected demand intercept for the risky major.

$$-E[a] = 1 - 2n_R - a_S$$

$$a_S - E[a] - 1 = -2n_R$$

$$\frac{1 + E[a] - a_S}{2} = n_R$$



The deviation, from 50-50, depends on how much higher expected demand is for the risky major relative to the safe major.

$$\frac{1 + a_S - E[a]}{2} = n_S$$

Wages are a little more interesting because the ex post realizations matter, and no indifference conditions need to hold ex post. Using the demand curves and the  $n_R$  above,

$$\begin{aligned} w_S &= \frac{a_S - (1 - n_R)}{b} = \frac{E[a] + a_S - 1}{2b}, \\ w_{R,H} &= \frac{a_H - n_R}{b} = \frac{a_S - 1 + a_H + (1 - p_H)(a_H - a_L)}{2b}, \\ w_{R,L} &= \frac{a_L - n_R}{b} = \frac{a_S - 1 + a_L - p_H(a_H - a_L)}{2b}, \end{aligned}$$

where  $E[a] = p_H a_H + (1 - p_H) a_L$ .

Under  $a_H > a_S > a_L$  and  $0 < p_H < 1$ , we have

$$w_{R,H} > w_S > w_{R,L}.$$

Ex post wages in the risky sector depend on realized demand, and the more confident people were ex ante of a bad state, the higher ex post wages are in the high state occur.

$$\frac{\partial w_{r,H}}{\partial p_H} = \frac{(a_L - a_H)}{2b} < 0$$

This is not a deviation from rational expectations, it is just that as  $p_H \rightarrow 0$ , entering a profession becomes more like a lottery ticket. This is the idea of a shortage of workers entering the risky profession, but wages rising ex post when the risk is not realized, (or - not modeled - if this is realized later than expected). A running example here is radiologists. This is the same idea, as in finance, there can be what looks like a “bubble” ex post, even with rational expectations.

## 6 Promise of this idea

- I find this question interesting, but I am likely racing against other people
- We are early for this to be studied empirically. The bend on the CS major curve seems to be a 2024-25 phenomenon. The current semester’s data will be released in January 2026
- Is this an interesting question if there is a null result? It seems more interesting if AI exposed majors were falling off a cliff, as that would be evidence people think this technology will impact employment and wages in the professions of that major.

Though, perhaps a null would be indicative that college students’ beliefs are that AI won’t really automate computer science jobs.

## References

Erik Brynjolfsson, Bharat Chandar, and Ruyu Chen. Canaries in the coal mine? six facts about the recent employment effects of artificial intelligence. Working paper, Stanford Digital Economy Lab, November 2025. URL <https://digitaleconomy.stanford.edu/publications/canaries-in-the-coal-mine/>.

Susan M Dynarski, Steven W Hemelt, and Joshua M Hyman. The missing manual: Using national student clearinghouse data to track postsecondary outcomes. Technical Report w19552, National Bureau of Economic Research, 2013.

## A College Major Data

### A.1 Three sources of measurement error [Dynarski et al., 2013]

**Matching Errors.** To get student-level data, NSC has to match its records to those from other data sources. So there will inevitably be match errors between those. **Roverage (both across and within universities)**

*Across universities:* The NSC does not get enrollment numbers from all universities. This means totals will not match the true number of college students. “As of fall 2011, the NSC reports that they cover 93 percent of postsecondary enrollment.” [Dynarski et al., 2013]. This number is up to 97% now as of 2025.

*Within universities:* “It is not necessarily reasonable to assume that NSC participating institutions report degree information for all of their students” **FERPA blockage** “FERPA is a federal law that protects the privacy of student education records. The law applies to all schools that receive funds from the U.S. Department of Education. Under FERPA, both students and schools can block their enrollment and degree information. So, NSC cannot release student-level information if the record is “FERPA-blocked.” ”

### A.2 RPS

### A.3 ACS

At least a 2-year lag. And doesn’t even include enrollments, as far as I can tell.

### A.4 IPEDS

Roughly a 1-year lag. “For example, the most recent data release on September 23, 2025, included provisional data from the Fall 2024 data collection” This is more comprehensive than

## A.5 National Survey of College Graduates

This data is biennial.